Descriptive Statistics Report

Data Analysis Assignment

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Dataset	Top Spotify Songs 2023
Column Analysed	stream
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Submitted in partial fulfilment of the Descriptive Statistics module

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1. Introduction

Write a short introduction (3–5 sentences). State what descriptive statistics are and why they are useful for analysing data.

Descriptive statistics help us to compile data in an easier and more effective way. We can determine the trends, distribution and patterns inside the dataset. This method helps to compare, clarify and read huge data with ease. We use different function to inspect different kinds of data's. This report contains descriptive statistics to investigate the most famous songs of 2023 on Spotify.

2. Dataset Description

Describe the dataset you were assigned. Include: the name of the dataset, its source (Kaggle link), the number of rows and columns, and the specific column you are analysing.

Dataset Name	Most streamed Spotify songs 2023
Source / Link	Kaggle
Number of Rows	943
Number of Columns	24
Column Analysed	Streams

Write a brief description of what the column measures and what type of data it contains (numerical / categorical.

__The dataset is called "Most Streamed Spotify Songs 2023" from kaggle. This dataset contains 943 rows and 24 columns. The column analyzed is Streams that represent how many times songs is played on Spotify. The columns as stream related data which has numerical values. The numerical data helps us to know the popularity of the songs and the greater the value the more popular the song is. ____

3. Frequency Table

Complete the frequency table below. Use the class intervals specified in your question. Add more rows if needed by right-clicking → Insert Row Below.

Class Interval	Frequency (f)	Relative Freq (%)	Cumulative Freq	Cumulative Rel Freq (%)
0-200M	356	37.36	356	37.36
200-400M	227	23.82	583	61.18
400-600M	103	10.81	686	71.98
600-800M	71	7.45	757	79.43
800-1B	44	4.62	801	84.05
1B-1.5B	82	8.60	883	92.65
1.5+B	70	7.35	953	100%
TOTAL	953	100%	953	100%

Observations from the Frequency Table

Write 3–4 sentences about what the frequency table shows. Which class interval has the highest frequency (modal class)? What does the cumulative relative frequency tell you?

The frequency table demonstrates that most songs are between 0-200 million. 37.36% of songs are in this interval. Which means most of the songs have not streamed much. The table shows that songs over 61% of songs have less than 400 million. Which clearly states that lot of songs do not get streamed much. As the number of stream increases, the number of the stream decreases, which shows fewer songs got popularity. The frequency table shows us the stream counts are unevenly spread out which shows distribution of songs is not fair.

4. Statistical Analysis

For each measure below, show your Python code output, state the calculated value, and explain what it means in the context of your dataset.

4.1 Mean

Paste or write the Python code you used to calculate the mean

```
import pandas as pd
df = pd.read_csv("spotify-2023.csv", encoding='latin1')
df['streams'] = pd.to_numeric(df['streams'], errors='coerce')
mean_value = df['streams'].mean()
print(f"mean_value:{mean_value:.2f}")
```

Calculated Mean: _______ 514137424.94________

Interpret the mean in one or two sentences. What does this average value tell you about the data?

___Mean is average number of stream per song which is 514 million streams. It show generally how popular the songs is.

4.2 Median

Paste or write the Python code you used to calculate the median.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("spotify-2023.csv", encoding='latin1')
df['streams'] = pd.to_numeric(df['streams'], errors='coerce')
median_value = df['streams'].median()
print(f"median_value:{median_value:.2f}")
```

Calculated Median: _________290530915.00________

Is the median higher or lower than the mean? What does the difference between them suggest about the shape of the distribution?

_The median is lower than the mean, where median is 290 million and mean is 514 million. This means the data's are unevenly distributed . The difference between mean and median shows that the number are skewed to the right. Moreover, the songs streams are really high for some of the songs which makes the mean higher this demonstrates that even though some most songs have average stream count but a smaller portion of songs have higher stream counts.

4.3 Mode

Paste or write the Python code you used to find the mode.

```
import pandas as pd
df = pd.read_csv("spotify-2023.csv", encoding='latin1')
df['streams'] = pd.to_numeric(df['streams'], errors='coerce')
mode_value = df['streams'].mode()[0]
print(f"mode_value:{mode_value:.2f}")
```

Calculated Mode: ___________ 156338624.00_____

Modal Class Interval (from frequency table): _______0-200M__________

Explain what the mode tells you. Does it match the modal class you identified in Section 3?

__________________The modal class interval is from 0-200M which matches perfectly as in the frequency table most songs from this interval.

4.4 Standard Deviation

Paste or write the Python code you used to calculate the standard deviation.

```
import pandas as pd
df = pd.read_csv("spotify-2023.csv", encoding='latin1')
df['streams'] = pd.to_numeric(df['streams'], errors='coerce')
std_value = df['streams'].std()
print(f"std_value:{std_value:.2f}")
```

Calculated Standard Deviation: _______ 566856949.04__________

What does the standard deviation tell you about the spread of the data? Calculate Mean ± 1 Std Dev and state how many values fall within this range.

_The standard deviation shows the distance between the stream value and mean. Here standard deviation is almost 566.86 million which means that there is a high distance between the mean and stream value whereas some songs have very high streams and some have very low streams. Almost of the stream values falls under 0 to 1.8 billion streams, so greater part of the songs falls under this range and minor part have very high streams counts.

4.5 Summary Statistics Table

Fill in the table below with all your calculated values and a brief interpretation of each.

Statistic	Calculated Value	What this tells us
Mean	514137424.94	Demonstrate average number of streams for each song.
Median	290,530,915.00	It displays the middle value
Mode	156,338,624.00	It shows the common stream.
Modal Class Interval	0-200M	It reveals most songs are in this range.
Standard Deviation	566,856,949.04	It illustrates the distance between mean and stream counts
Range (Max − Min)	3,703,892,312	It reveals the difference between highest streams to lowest streams
Minimum Value	2,762.00	It shows the lowest streams
Maximum Value	3,703,895,074.00	Shows the highest streams

5. Charts and Visualisations

Insert your charts below. Right-click the grey box → 'Insert Picture' (or copy-paste from your notebook). Write a caption and short description for each chart.

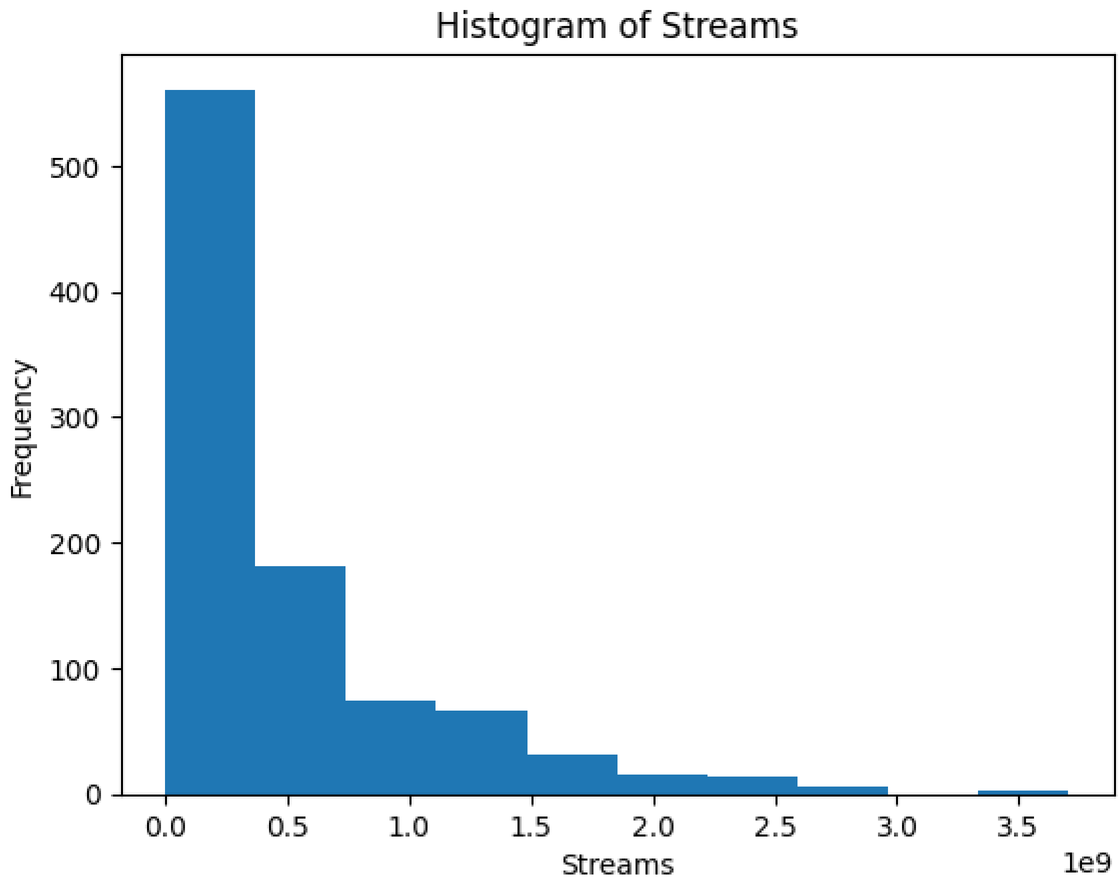
5.1 Chart 1 — Histogram / Bar Chart

Insert your primary chart (histogram or bar chart as specified in your question).

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("spotify-2023.csv", encoding='latin1')
df['streams'] = pd.to_numeric(df['streams'], errors='coerce')
plt.hist(df['streams'], bins=10)
plt.xlabel("Streams")
plt.ylabel("Frequency")
plt.title("Histogram of Streams")
plt.show()
```

Figure 1: ____________



____

Describe what this chart shows. Where is the peak? Is the distribution symmetric or skewed?

___The histogram demonstrates the streaming counts of the songs. Here we can notice the bar is peak from 0 to 200M which shows most songs have lowest stream counts. The distribution is right skewed.as tail is towards the high streaming values. Which most songs have low stream counts____

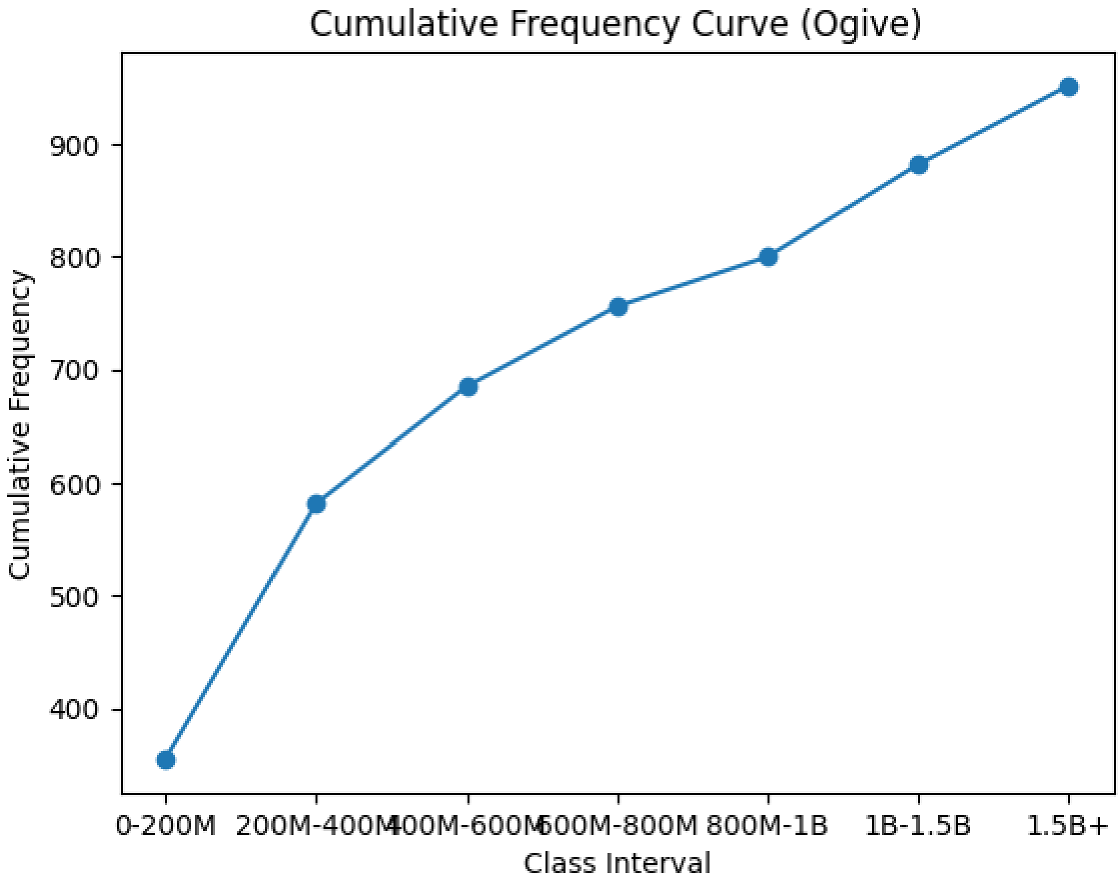
5.2 Chart 2 — Additional Chart

Insert your second chart if required (e.g., bar chart of top values, pie chart, ogive/cumulative frequency chart).

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("spotify-2023.csv", encoding='latin1')
df['streams'] = pd.to_numeric(df['streams'], errors='coerce')
bins = [0, 200000000, 400000000, 600000000, 800000000, 1000000000,
1500000000, df['streams'].max()]
labels = ['0-200M','200M-400M','400M-600M','600M-800M','800M-1B','1B-
1.5B','1.5B+']
df['class'] = pd.cut(df['streams'], bins=bins, labels=labels)
freq = df['class'].value_counts().sort_index()
cum_freq = freq.cumsum()
plt.plot(labels, cum_freq, marker='o')
plt.xlabel("Class Interval")
plt.ylabel("Cumulative Frequency")
plt.title("Cumulative Frequency Curve (Ogive)")
plt.show()
```

Figure 2: _____



Describe what this chart shows and how it relates to the statistics you calculated.

The cumulative frequency graph indicates how the number of songs increases at different stream intervals. The curve is increasing steeply in the lower stream range up to 400M and then the curve becomes less steeply after that showing the songs have high streams. The sequence supports the statistical result where the median is lower than the mean which shows right skewed distribution. It also matches with histogram and standard deviation which indicates most songs have lower streams and less songs have higher streams.

6. Discussion and Interpretation

Write a full discussion (at least 2 paragraphs). Answer ALL of the interpretation questions from your assignment question. Use your calculated values to support your points.

Mean vs. Median — What does the difference tell us?

_Here the mean is higher than the median the value of mean is (514137424.94) and the value of median is (290,530,915.00). Which states the distribution of the graph is positively skewed. Which shows small number of songs have high streams and huge number of songs have lower streams. The histogram is also in the favors of this because most of the songs comes under the lower ranges

Standard Deviation — What does the spread tell us?

__The value of standard deviation is 566,856,949.04. Which mean there is a high gap between the stream values and mean which indicates that that is a high level of variation in songs popularity. Some songs have low stream whereas other has extremely high values creating a huge gap in between. This spread tells us that the dataset is highly variable and unevenly distributed.

Outliers (if any)

List any values more than 2 standard deviations above or below the mean. Are they genuine outliers or data errors?

________Outliers are values that are much higher or lower than most of the data. In this dataset, songs with streams above about 1.65 billion can be considered outliers. These songs are extremely popular compared to others. They are not errors, but real songs with very high popularity.

7. Conclusion

Write a conclusion (3–5 sentences). Summarise your key findings. Which single statistic best describes the 'typical' value in your dataset — the mean or the median — and why?

In conclusion, the analysis shows that most songs have low to moderate stream counts, while only a few songs achieve extremely high popularity. The distribution is positively skewed, as shown by the higher mean compared to the median. The standard deviation also indicates a large variation in the data. Among all measures, the median is the best representation of a typical song because it is not affected by extreme values like the mean.

8. References

List all sources you used, including the dataset source and any textbooks or websites referenced.

https://www.w3schools.com/python/matplotlib_intro.asp
https://www.w3schools.com/python/python_file_handling.asp